Statistical Inference and Modeling

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Lectures

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Appendix. Resampling Methods

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Overview of resampling methods

- A process of repeatedly drawing samples from a training set and refitting a model on each sample.
- We seek for information that would not be obtained from fitting the model only once using the original training sample.
- **Cross-validation**: used in estimation of test error or model flexibility.
- **Bootstrap**: a measure of accuracy of a parameter estimate.
- Resampling approaches can be **computationally expensive** but with nowadays technology, it becomes less prohibitive.

Resampling methods: bootstrap

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Bootstrap

Bootstrap is a scheme of estimating an unknown statistical parameters by **repeatedly** *sampling with* **replacement** *from the original data set.*

Steps.

- Prepare the data sampling for N trials
- For each trial, sample m data points to calculate a statistical value (*e.g.*, $\hat{\alpha}^i$)
- Repeat the above steps until you compute all $\hat{\alpha}^1, \hat{\alpha}^2, ..., \hat{\alpha}^N$.
- Plot histogram of $\hat{\alpha}^1, \hat{\alpha}^2, ..., \hat{\alpha}^N$, and use the histogram to justify the most likely $\hat{\alpha}$.

Example

Problem settings

- Suppose we have to devide the investment budgets for two companies. We want to know α, which is a ratio of investment budgets.
- In order to know the most appropriate α, we collect the yield returns for each investment denoted as X and Y.
- Thus, our totoal yield return will be the combination between the ratio of the two investment yields, that is,

$$Z = \alpha X + (1 - \alpha)Y$$

Our goal is to ensure the stable yield return. That is, we want to find α minimize

$$\hat{\alpha} = \underset{\alpha}{\textbf{argmin}}\ \text{Var}[\alpha X + (1 - \alpha)Y]$$

- One can show that the solution α that minimize the variance is given by

$$\hat{\alpha} = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

Why we need **boostrapping**?

- Boostrapping is needed to be used in calculating the statistical values σ_Y^2, σ_X^2, and σ_{XY}.

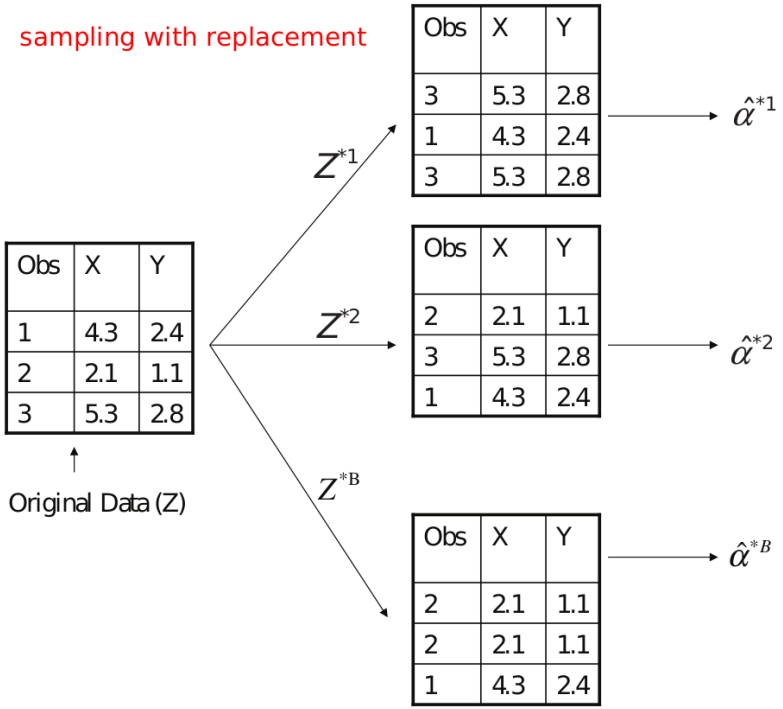
Example

Steps

- Prepare the data sampling for N ($N = 1000$) trials.
- For each trial, sample m data points to calculate a statistical value $\hat{\sigma}_Y^2, \hat{\sigma}_X^2$, and $\hat{\sigma}XY$.
- For each trial, compute $\hat{\alpha}^i$ using $\hat{\sigma}_Y^2, \hat{\sigma}_X^2$, and $\hat{\sigma}XY$.

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}XY}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\sigma_{XY}}$$

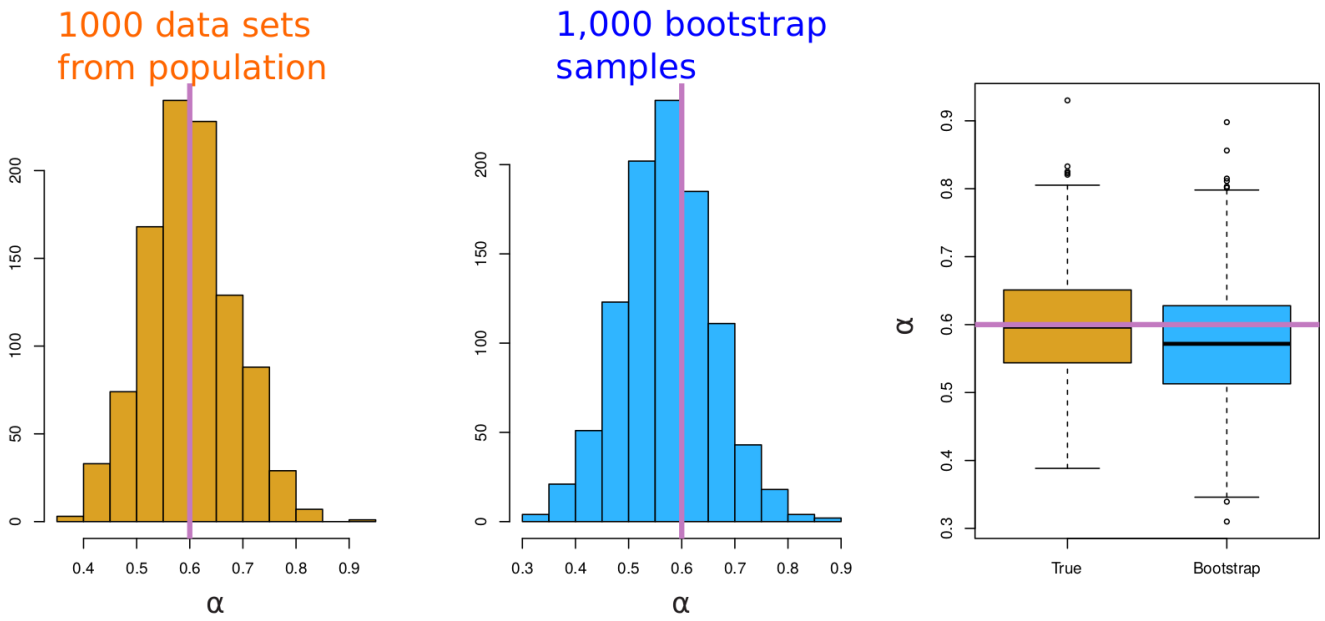
- Plot histogram of $\hat{\alpha}^1, \hat{\alpha}^2, ..., \hat{\alpha}^N$, and use the histogram to justify the most likely $\hat{\alpha}$.



Use each of new sampled dataset to compute a new estimate (such as α).

Example (cont')

Here, we compare the bootstrapped $\hat{\alpha}$ (Blue) with the true sampled values of α (Yellow).



- Histograms of $\hat{\alpha}$ from two approaches are similar and the sample means are close.
- Standard deviations of $\hat{\alpha}$ are 0.083 (1,000 datasets) and 0.087 (bootstrap).
- The box plots of $\hat{\alpha}$ are also quite similar (true α is 0.6).
- We can use bootstrap when we cannot generate new samples from population.

References I

[1] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY, USA: Springer New York Inc., 2001.

[2] Gareth James et al. *An Introduction to Statistical Learning: with Applications in R*. Springer, 2013. URL: https://faculty.marshall.usc.edu/gareth-james/ISL/.

[3] Jitkomut Songsiri. "Statistical Learning". In: in EE575 Random Processes for EE. Chulalongkorn University, 2022. Chap. 5. Resampling. URL: http://jitkomut.eng.chula.ac.th/ee575.html.